

A2 Computer Science — Python Practice Set

Avoid Python shortcuts (`sum()`, `sorted()`, slicing tricks) in your first pass through each question, since the exam expects you to demonstrate the underlying logic (loops, counters, comparisons) manually.

String Handling

1. Write a program that takes a string input and outputs:
 - its length
 - the string in uppercase and lowercase
 - the string reversed
2. Write a program that counts the number of vowels, consonants, digits, and spaces in a given string.
3. Write a program that takes a sentence and capitalises the first letter of every word (without using `.title()`).
4. A string `PlateNumber` stores a vehicle registration in the format "ABC-1234" (3 letters, hyphen, 4 digits). Write code that validates whether a given string matches this exact format, checking character by character (do not use regular expressions).
5. Write code that removes repeated consecutive characters from a string (e.g. `"aaabccddd" → "abcd"`).
6. Ask the user to enter a word. Output the first character, the last character, and the total number of characters in the word.
7. Ask the user to enter a word and count how many times the letter "a" appears in the word.
8. Ask the user to enter a word and a character they want to replace. Then ask for the new character and output the updated word.

1D Array

1. Write a Python program to store 5 student names in a 1D array. Then ask the user to enter an index and output the name stored at that index.
2. Write a Python program to input 5 numbers from the user and store them in a 1D array. After all the numbers have been entered, use a loop to calculate and output the total of all the numbers.
3. Write a program that declares a 1D array of 10 integers, populates it with values entered by the user, then outputs the sum and average.
4. A 1D array holds 20 exam scores (0–100). Write a program that outputs:
 - the highest and lowest score
 - the average score
 - how many students scored above the average
5. Declare an array `Temperatures` of 7 real numbers representing daily temperatures for a week. Write an algorithm that:
 - outputs the day number (1–7) with the highest temperature
 - calculates how many days had a temperature above 25.0
 - reverses the array in place (without using `reverse()`)